

A5 · Errors on Purpose

CS425 Computer Networks · instructor led · every command runs on Onyx (ssh onyx)

TURN THIS IN ON PAPER
before you leave the room
graded pass/fail

Names _____ Date _____

Every round: **predict** first, in ink, then **run** the command with the projector, then **check**. A wrong prediction costs nothing. A prediction written afterwards is not a prediction.

ROUND 1 · IS ANYBODY OUT THERE? ping · curl

QUESTION	ANSWER
Predict: will ping -c 3 8.8.8.8 get replies?	
What did ping 8.8.8.8 report?	
Did the gateway 132.178.227.1 answer? What did curl print?	

Q1. So what is actually broken? What do you now know about how campus treats ICMP echo, compared with TCP on port 443?

ROUND 2 · TRACEROUTE BY HAND ping -c 1 -t TTL 8.8.8.8

TTL	PREDICT: WHAT COMES BACK, AND FROM WHOM	WHAT ACTUALLY CAME BACK
-t 1		
-t 2		
-t 3		

Q2. Who sent the Time to live exceeded, and how do you know from the output alone? Nothing came back at -t 2 and -t 3. Does that mean there is no router at hop 2 or hop 3?

ROUND 3 · HOW DO YOU KNOW YOU ARRIVED? UDP to 198.60.22.13 port 33434

QUESTION	ANSWER
Predict: a reply, a 3 second timeout, or something else?	
What did Python print? And for 127.0.0.1?	

Q3. UDP has no connection, so what was refused, and who said so? Which ICMP message came back, and how did it become an error in a Python program?

ROUND 4 · SAME PATH, THREE WAYS `mtr -n [-u | -T -P 443] -r -c 5 mirrors.xmission.com`

PROBE	PREDICT: HOW FAR WILL THE TRACE GET?	LAST HOP THAT ANSWERED	HOP 2'S ADDRESS, IF ANY
ICMP echo (default)			
UDP -u			
TCP -T -P 443			

Q4. Which hop do you think is throwing away your pings? What is your evidence? What kind of address does hop 2 have (think back to A4, round 1)?

Q5. Hops 6 to 9 lose 100% of the probes, yet hop 13 loses none. How can a router drop every probe and still forward all of your traffic?

Exit. A friend runs `ping 8.8.8.8` on Onyx, sees 100% packet loss, and says Onyx is off the Internet. Give them two pieces of evidence from today that say otherwise.

Graded pass/fail on something in every box. Hand this in before you leave.